Quiz 1 Solution:

How is the Requirement Traceability Matrix used in software testing life cycle?

Answer to Question 1:

- Creation of RTM at requirement analysis phase of STLC ,
- Test Development phase uses this document to create the test cases, so that each of the test cases can be traced back to each of the requirements
- Test results are updated in RTM so that the bugs can be mapped to any of the requirements in the document.

Explain how to verify the High Level Design of any system.

Answer to Question 2:

The tester verifies the high-level design. All the interfaces and interactions of the user/customer (or any person who is interfacing with the system) are specified in this phase. The tester verifies that all the components and their interfaces are in tune with the requirements of the user. Every requirement in SRS should map to the design.

Three types of design documents are verified:

- Interface design
- Data Design
- Architectural Design

You have a login module with a password field that can not be shorter than 8 and longer than 28(including) characters. This password should contain at least one numeric and at least one capital letter. Calculate how many test cases can be created for BVC, Robust, Worst-case testing method. What boundary values will be used for these kinds of testing?

Answer to Question 3:

8<=Length of Password<=28

1<=No. of Numeric<=27

1<=No. of Capital Letter<=27

3 Variables, so for BVC no. of test cases 4*3+1=13

For Robust no. of test cases 6*3+1=19

For worst case 5^3 =125

Label	Len of Password	No. of Numeric	No. of Capital
Min+	9	2	2
Min	8	1	1
Min-	7	0	0

Nom	15-18(Any number)	12-14 Any Number	12-14 Any Number
Max+	29	28	28
Max	28	27	27
Max-	27	26	26

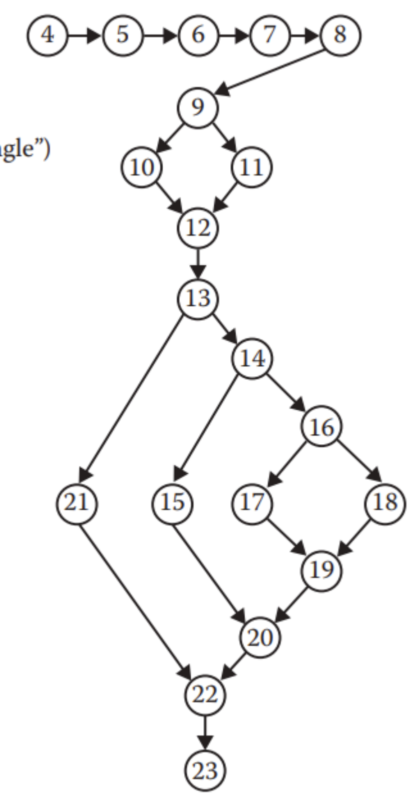
Explain the goals an equivalence partitioning method should maintain.
Answer to Question 4:

- Completeness: Without executing all the test cases, we strive to touch on the completeness of the testing domain.
- Non-redundancy: When the test cases are executed with inputs, then there is redundancy in executing the test cases. Time and resources are wasted in executing these redundant test cases, as they explore the same type of bug.

Quiz 2 Solution:

Answer to Question No 1:

```
1  Program triangle2
2  Dim a,b,c As Integer
3  Dim IsATrinagle As Boolean
4  Output("Enter 3 integers which are sides of a triangle")
5  Input(a,b,c)
6  Output("Side A is", a)
7  Output("Side B is", b)
8  Output("Side C is", c)
9  If (a < b + c) AND (b < a + c) AND (c < a + b)
10    Then IsATriangle = True
11    Else IsATriangle = False
12 EndIf
13 If IsATriangle
14    Then  If (a = b) AND (b = c)
15             Then Output ("Equilateral")
16             Else  If (a≠b) AND (a≠c) AND (b≠c)
17                      Then Output ("Scalene")
18                      Else Output ("Isosceles")
19                   EndIf
20             EndIf
21   Else  Output("Nota a Triangle")
22 EndIf
23 End triangle2
```



Calculate the cyclomatic complexity for the graph in Question 1.
Answer to Question 2:
Using any equation: Complexity would be 5
E-N+2*P= 23-20+2*1=5
D+P= 4+1 =5
No. of regions = 5

Teachers must apply for the incentive after 2010. Teachers who are currently eligible to retire (age ≥ 63 years) shall have a multiplier of 1.6% on their salary up to, and including, $90,000. Teachers who meet the 80 total years of age plus years of teaching shall have a multiplier of 1.55% on their salary up to, and including, $90,000.
Make a decision table to describe the retirement pension policy; be sure to consider the retirement eligibility criteria carefully.
Answer to Question 3:

Stubs	Rule 1	Rule 2	Rule 3	Rule 4	Rule 5
age>=63	I	T	T	T	F
age+teaching_age>=80	I	F	T	T	F
application_year>=2010	F	T	T	T	T
1.6% of Salary upto 90,000$		X		X	
1.55% of Salary upto 90,000$			X	X	
No incentive	X				X